

Regional Sales Analysis with DataCo

C1 · From W1 to defensible evidence

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1. Problem, user and decision

Selected commercial line from W1

Research question

How do the best-selling product categories vary across regions, and what factors are associated with lower sales volumes?

Proposed user

Commercial analyst

Decision

Which region–category combinations should be prioritized for further analysis before making assortment or promotional decisions?

- Candidate A (sales) was retained over Candidate B (logistics) because it better fits the commercial variables and the intended decision.
- The analysis is historical and descriptive: associations are not interpreted as causal effects.

2. Seven-step analytical framework

A direct mapping from the business decision to measurable evidence

1 • Decision

Prioritize **region–category combinations** for further commercial analysis

2 • Success

Identify **meaningful and comparable sales patterns** that can support commercial prioritization.

3 • Questions

Compare **sales volume, category share, units per order and discount patterns** across regions.

4 • Process

Order → item → status → shipment

5 • Entities

Customers, orders, items, states

6 • Representation

1 row = 1 order item

7 • Measurement

Define the KPIs used to compare regions and categories. **Main KPI: Units per Order**, complemented by Recorded Units and Category Share.

3. Data preparation and the Five Cs

DataCo Supply Chain · 180,519 order-item records · 53 variables

Observation unit

1 row = 1 Order Item
Order Item Id is unique.

Commercial subset

172,765 retained items
62,897 represented orders

Status exclusions

3,692 CANCELED
4,062 SUSPECTED_FRAUD

Clean

No duplicate item IDs; essential fields validated.

Consistent

Temporal coverage differs across regions/categories.

Conformed

No external join; category ID/name consistency checked.

Current

2015–2018: suitable for historical analysis, not 2026 operations.

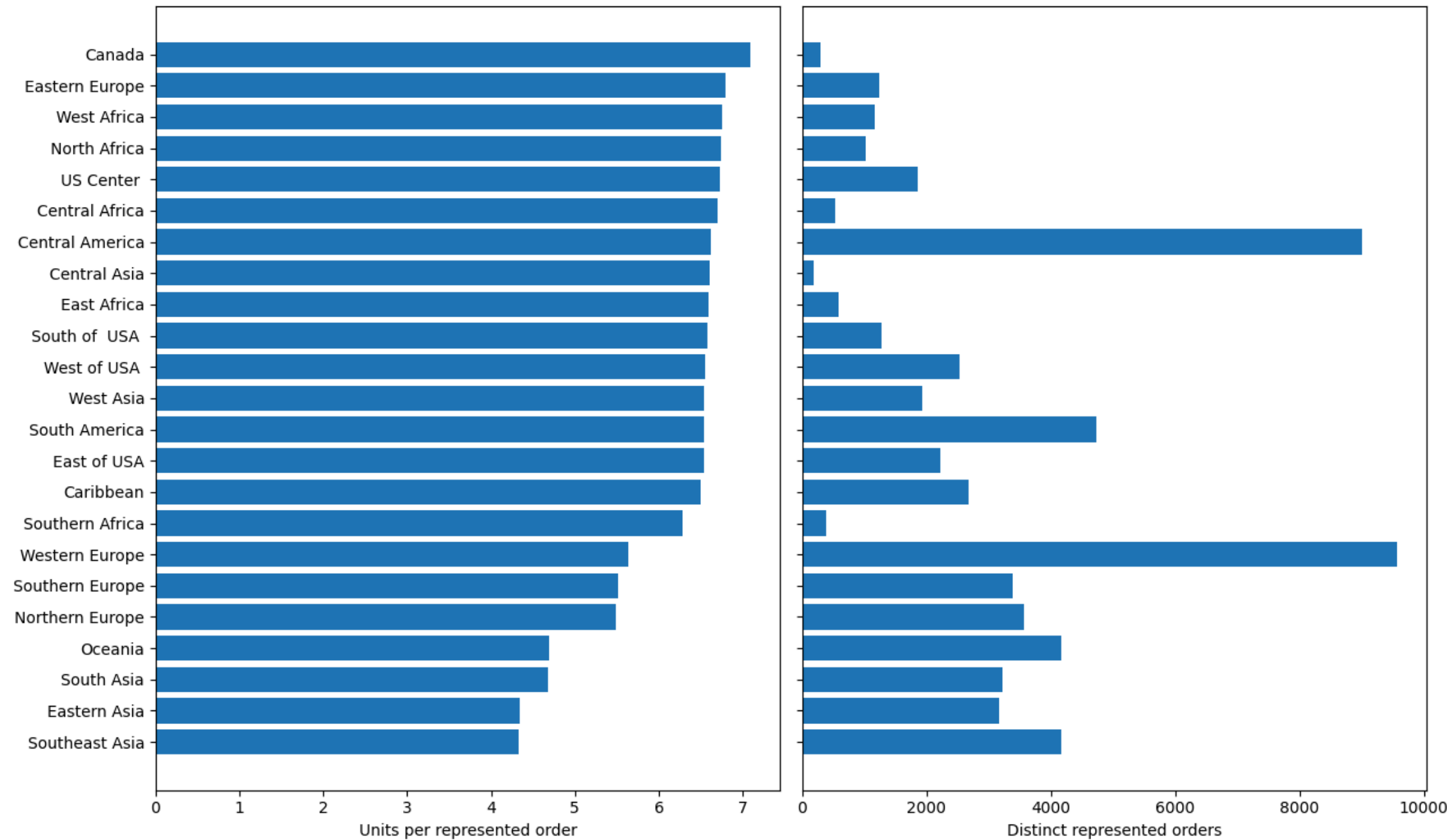
Comprehensive

No stock, unmet demand, campaign exposure or competitor data.

4. EDA — Regional volume is not regional performance

Raw volume and normalized purchasing intensity tell different stories

Purchase intensity and sample size — eligible records



Central America

59,540 retained units

Canada

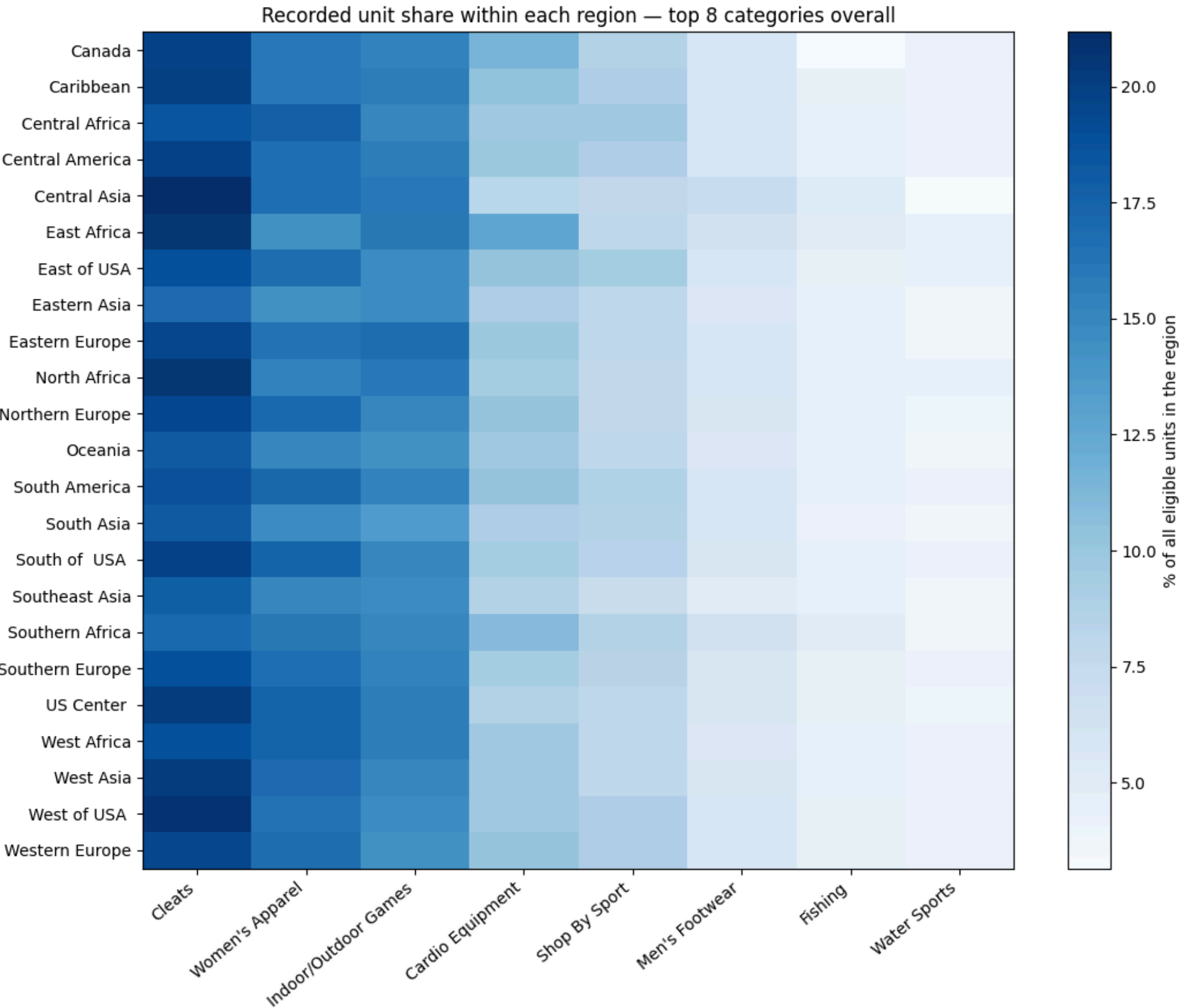
2,063 retained units
7.09 units per represented order

Interpretation

Whole-period rankings mix geography, assortment and time coverage. They describe the dataset; they do not prove regional performance.

EDA visual — Category mix across regions

Recorded unit share within each region for the eight largest categories overall



What the chart shows

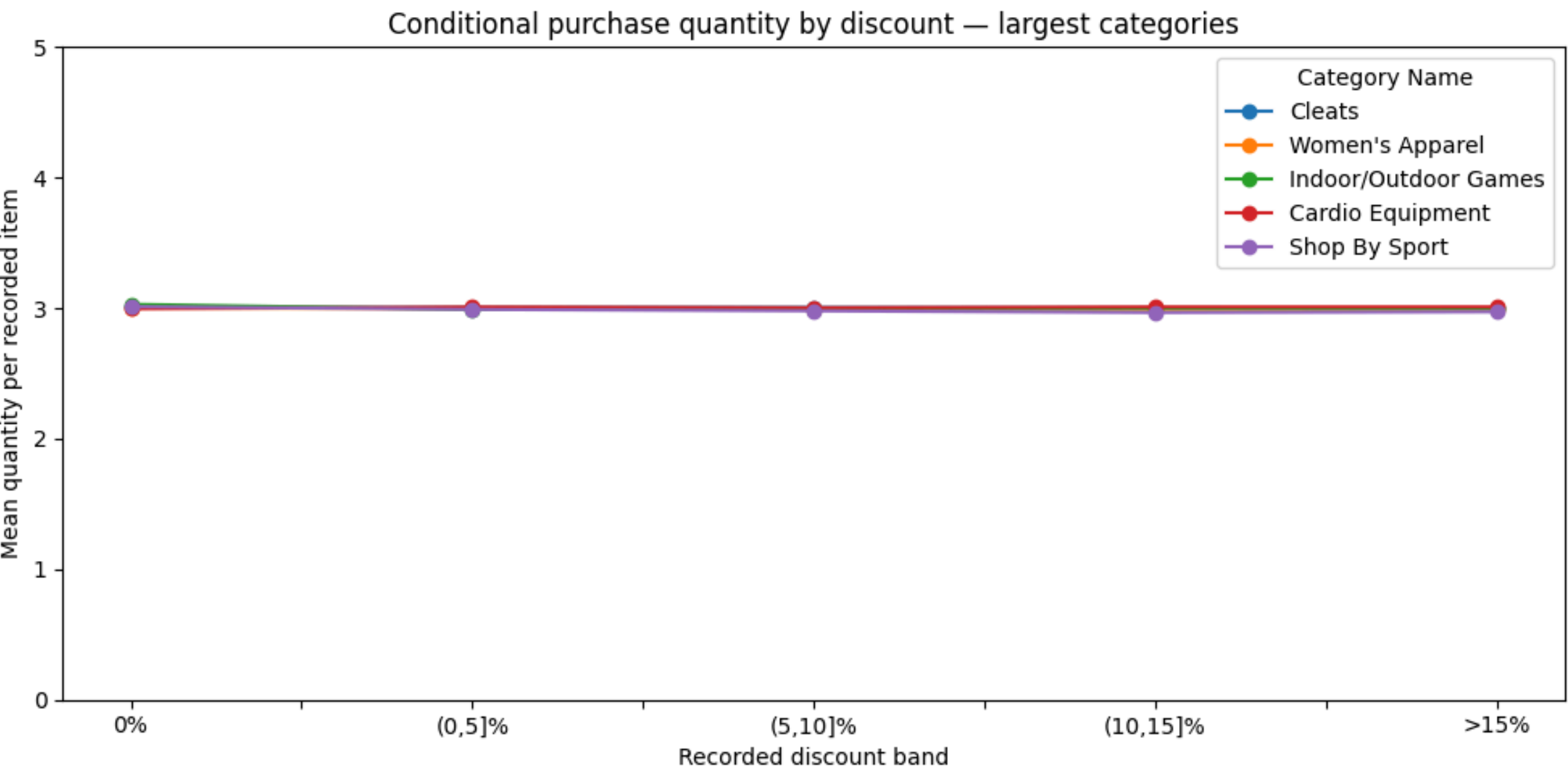
Category composition is relatively similar across many regions: Cleats, Women's Apparel and Indoor/Outdoor Games account for large shares of recorded units.

Interpretation boundary

These shares describe the observed DataCo transactions. They do not measure total regional demand or market potential.

EDA visual — Discount bands show little separation

Conditional purchase quantity remains close to three units per recorded item across the largest categories



Observed pattern

Across the five largest categories, the max–min difference in mean item quantity ranges from only 0.008 to 0.045 units.

Commercial implication

The descriptive evidence does not justify recommending deeper discounts. Causal effects are not identified.

5. Robustness checks and discount analysis

The main descriptive findings are stable, but they do not support causal recommendations

Status sensitivity

Largest absolute change in regional category share after excluding canceled/fraud items:
≈ 0.54 percentage points

Discount bands

Across the five largest categories, max–min mean item quantity differs by only:
0.008–0.045 units

Commercial implication

The observed discount pattern does not justify recommending deeper discounts.

- Monetary arithmetic checks found no discrepancies above 0.05 in the retained subset.
- Lower recorded sales are not proof of weak underlying demand.
- Alternative explanations include assortment, product availability, regional activity, order states and unequal observation windows.

6. Main limitation — unequal temporal coverage

Comparability is the central constraint on the regional interpretation

Material finding

Canada is observed only from Aug 2016 to Jan 2017, while other regions have different observation windows.

Why it matters

Units per order reduces scale differences, but it does not remove time or assortment confounding.

Required next check

Restrict future comparisons to overlapping periods and available categories, and report the restricted sample.

- Use volume, category share, units per order and sample size jointly.
- Review reporting coverage and assortment before making investment decisions.
- Seek inventory, campaign exposure/cost, margins and customer opportunity data before prescribing action.

7. Project continuity — ML and dashboard

C1 establishes feasibility; implementation comes later

Future ML task

Unit: region–category–month

Target: next month's recorded units

Potential inputs: lagged quantities, prior order counts, category shares and calendar variables.

Risk: avoid leakage from information only known after the prediction point.

Dashboard proposal

User: commercial analyst / category manager

KPIs: recorded units, orders, units/order, category shares

Filters: month, region, category, customer segment, status inclusion

Show sample sizes and coverage warnings.

Decision boundary

The dashboard should support investigation, not automatically prescribe discounts or assortment changes.

If temporal coverage remains too sparse, clustering region–category profiles is a feasible alternative.

8. Conclusions

What the evidence supports — and what it does not

- DataCo supports a defensible descriptive comparison of regional and category sales patterns.
- Central America has the highest retained recorded volume, while Canada shows high units per represented order on a much smaller and shorter sample.
- Status exclusions have limited impact on category-share composition; discount-band differences are small.
- Unequal temporal and assortment coverage prevents interpreting whole-period rankings as comparable regional performance.
- Recommended use: shortlist region–category combinations for deeper investigation, then add inventory, campaign, margin and opportunity data before action.